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Privacy-preserving activity aggregation mechanism

Abstract

This disclosure relates to a method for privacy-preserving web activity monitoring including receiving, from an application on a user device of a user, a request for digital content from a domain, assigning, to the application and at a first time, a randomized cohort constructed based on a randomly selected identifier and a timestamp indicating the first time at which the randomized cohort was assigned to the application, and providing, to the application and at the first time, (i) a digitally signed certificate corresponding to the randomly selected identifier and the timestamp and (ii) a unique public key and corresponding unique private key associated with the certificate, wherein the randomly selected identifier is also assigned to at least a threshold number of other applications executing on other user devices within a predetermined period of time of the assignment of the randomized cohort to the application.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage Application under 35 U.S.C. § 371 and claims the benefit of International Application No. PCT/US2021/020694, filed Mar. 3, 2021, the entirety of which is incorporated herein by reference.

TECHNICAL FIELD

(2) This specification relates to web activity aggregation, data processing, and protecting user privacy in an online environment. The enhancement of online user privacy has led many browser developers to change the ways in which user data is handled. For example, some types of cookies are no longer being supported by some browsers, but the deprecation of third party (3P) cookies may lead to fraud and abuse.

BACKGROUND

(3) Aggregating web activity allows personalization of the browsing experience for a user, and enables delivery of content that is more relevant to a user, faster, than without monitoring. However, existing mechanisms such as cookies can be linked to a single user and information about the user. Such precision can make users feel like they can be too easily identified and their information is too easily compromised.

SUMMARY

(4) In general, one innovative aspect of the subject matter described in this specification can be embodiment in a method for privacy-preserving web activity monitoring that includes receiving, from an application on a user device of a user, a request for digital content from a domain, assigning, to the application and at a first time, a randomized cohort constructed based on a randomly selected identifier and a timestamp indicating the first time at which the randomized cohort was assigned to the application, and providing, to the application and at the first time, (i) a

digitally signed certificate corresponding to the randomly selected identifier and the timestamp and

(ii) a unique public key and corresponding unique private key associated with the certificate, wherein the randomly selected identifier is also assigned to at least a threshold number of other applications executing on other user devices within a predetermined period of time of the assignment of the randomized cohort to the application.

(5) In some implementations, the method includes receiving, from the application, a second request for digital content from the domain, and providing, by the application and to the domain at a second time, a vague identifier corresponding to the randomly selected identifier and a randomized cohort age bucket that indicates a range of ages of cookies that contains the age of the randomized cohort, wherein the age of the randomized cohort is calculated based on the difference between the second time and the first time.

(6) In some implementations, the method further includes detecting, by the domain and based on the received randomized cohort age bucket, abnormal activity associated with the randomly selected identifier and at least one of: a number of interactions associated with the randomly selected identifier, a randomized cohort age distribution, and a probability distribution associated with a particular interaction and a particular period of time.

(7) In some implementations, wherein assigning, to the application, a randomized cohort includes assigning, by the domain, the randomized cohort to the application, and wherein the randomly selected identifier is assigned to the at least a threshold number of other applications, wherein the randomly selected identifier is a randomly generated identifier selected from among two or more randomly generated identifiers, and wherein the unique public key is generated by the domain.

(8) In some implementations, assigning, to the browser, a randomized cohort includes assigning, by a central server, the randomized cohort to the application, wherein the randomly selected identifier is assigned to the at least a threshold number of other applications, wherein the randomly selected identifier is a randomly generated identifier selected from among two or more randomly generated identifiers, and wherein the unique public key is generated by the central server.

(9) In some implementations, the method includes providing, by the application, a request for digital content from a second domain different from the first domain and the randomized cohort, in response to providing the request for digital content from the second domain, receiving, by the application and from the second domain, an attestation request that includes a challenge, and providing, by the application and to a verification system, the digitally signed certificate, which triggers the verification system to (i) create a vague certification comprising the randomly selected identifier, the randomized cohort age bucket, and the challenge, (ii) sign the vague certification, and (iii) provide the vague certification to the second domain, wherein the challenge is blinded from the verification system using a blinding scheme.

(10) In some implementations, the method includes providing, by the application and to a verification system, the digitally signed certificate, and verifying, by the verification system, that the randomized cohort is assigned to at least a threshold number of people.

(11) Other embodiments of this aspect include corresponding systems, apparatus, and computer programs, configured to perform the actions of the methods, encoded on computer storage devices.

(12) The subject matter described in this specification can be implemented in particular embodiments so as to realize one or more of the following advantages.

(13) The ways in which a digital component distribution system selects and distributes personalized digital components (e.g., generate selection parameters and/or the selection parameters themselves) have historically included using user information (e.g., browsing information, interest group information, etc.) obtained from third-party cookies, which are cookies dropped on the client device by a domain (e.g., eTLD+1) that differs from the domain of the web page being rendered on the client device. However, some browsers are blocking the use of third party cookies, making it more difficult to select and provide personalized digital components, meaning that computing resources and bandwidth may be wasted by selected and distributing content to users that is not of

interest to the users. Furthermore, features that a computer system could previously perform using the third-party cookies are no longer able to be performed, thereby resulting in a less efficient and less effective computer system. To overcome this problem, privacy preserving techniques that enable the monitoring, aggregation, and analysis of web activity, while impeding the tracking of users, and while preventing the leakage of user information across computing systems can be used. In other words, the techniques discussed herein are changing the way that the computing systems operate to overcome problems that arise when browsers do not support the use of third-party cookies.

(14) The privacy-preserving monitoring mechanism described herein, randomized cohorts, provides web activity monitoring functionality. Randomized cohorts include an identifier and a timestamp. The combination of the identifier and the timestamp cannot uniquely identify a particular browser or user device, and instead. However, the timestamp can be obfuscated while still providing useful information by generating an age bucket to which the randomized cohort belongs and providing the combination of the identifier and the age bucket. The identifier and age bucket combinations are assigned to at least a threshold number of unique browsers running on different user devices, thus guaranteeing anonymity without sacrificing statistical utility of the randomized cohort. Randomized cohorts can be used to generate statistics regarding the activity of the cohort and to other information, while ensuring user anonymity. Users are randomly grouped into cohorts of size k , such that a domain to which the randomized cohort is scoped can track the activity of the cohort but not the activity of any one user.

(15) Additionally, randomized cohorts can be used in a security context for third parties, such as content providers and hosts, to detect fraudulent activity or coordinated misuse. For example, randomized cohorts allow for existing counter-abuse techniques to combat engagement abuse. Engagement abuse can include behavior such as click fraud, view count inflation, rating manipulation, rank manipulation, etc. Randomized cohorts can be used to detect suspicious web activity indicating fraudulent usage while providing users with a specific level of privacy that has not previously been available to users. For example, randomized cohorts can be used to provide users with a k -anonymity guarantee. A guarantee of k -anonymity ensures that there are at least k random users associated with a single randomized cohort, which can be identified by a randomized cohort identifier and a timestamp. For example, a guarantee of k -anonymity for a randomized cohort identifier where $k=100$ ensures that there are at least 100 random users associated with the randomized cohort identifier, such that information associated with a particular randomized cohort is anonymized to certain extent while remaining useful for applications such as statistical analysis and abuse detection.

(16) The described monitoring mechanism, randomized cohorts, improves user experience and trust by providing privacy guarantees that can be externally verified by independent third parties. The described system can include one or more verification servers that are independent of the sources of the randomized cohorts, such that the randomized cohorts and specific identity of a user remains hidden while allowing the verification server to determine the statistical properties of a particular randomized cohort identifier. This allows users to confirm, through an independent source, that their anonymity is being maintained and that the privacy-preserving system is functioning as promised. Users who can individually substantiate guarantees of privacy may feel more comfortable adopting a system that uses the described monitoring mechanism.

(17) Furthermore, randomized cohorts may be used as a replacement in systems using traditional methods of web activity monitoring. For example, randomized cohorts can be used in existing systems with little adaption required under certain conditions, allowing system designers to re-use existing infrastructures for providing relevant statistical data regarding web activity and performing security functions to protect third parties while improving user privacy.

(18) The techniques discussed throughout this document can also be used to detect irregular activity (e.g., network attacks), and shut down the irregular activity. For example, these techniques

can detect a higher than usual level of network requests or traffic, and use that information to prevent further network requests or traffic, or blocking further requests from a group of computing devices that are responsible for the high level of network requests or traffic. The techniques can also be used to detect overuse of specific computing resources, and perform load balancing to improve the efficiency of a computer system.

(19) Various features and advantages of the foregoing subject matter is described below with respect to the figures. Additional features and advantages are apparent from the subject matter described herein and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram of a system in which a privacy-preserving monitoring mechanism is implemented.

(2) FIG. 2 is a data flow diagram of an example process for issuing and implementing a privacy-preserving monitoring mechanism.

(3) FIG. 3A is a swim lane diagram that illustrates an example process for issuing and implementing a privacy-preserving monitoring mechanism that is limited in scope to a single issuing domain.

(4) FIG. 3B is a swim lane diagram that illustrates an example process for issuing and implementing a privacy-preserving monitoring mechanism that is used across web activity performed across different domains.

(5) FIG. 4 is a flow diagram that illustrates an example process for issuing and implementing a privacy-preserving monitoring mechanism.

(6) FIG. 5 is a block diagram of an example computer system.

(7) Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

(8) In general, this document describes systems and techniques for guaranteeing a specified level of privacy for users associated with a monitoring mechanism, a randomized cohort, that also provides statistical tracking and abuse detection capabilities.

(9) FIG. 1 is a block diagram of an environment **100** in which for privacy preserving data collection and analysis. The example environment **100** includes a network **102**, such as a local area network (LAN), a wide area network (WAN), the Internet, or a combination thereof. The network **102** connects electronic document servers **104** (“electronic doc servers”), user devices **106**, a digital component distribution system **110** (also referred to as DCDS **110**), one or more verification servers **130**. The example environment **100** may include many different electronic document servers **104**, user devices **106**, verification servers **130**, and trusted domain servers **140**. For ease of explanation, one trusted domain server **140** is shown.

(10) A user device **106** is an electronic device that is capable of requesting and receiving resources (e.g., electronic documents) over the network **102**. Example user devices **106** include personal computers, wearable devices, smart speakers, tablet devices, mobile communication devices (e.g., smart phones), smart appliances, and other devices that can send and receive data over the network **102**. In some implementations, the user device can include a speaker that outputs audible information to a user, and a microphone that accepts audible input (e.g., spoken word input) from the user. The user device can also include a digital assistant that provides an interactive voice interface for submitting input and/or receiving output provided responsive to the input. The user device can also include a display to present visual information (e.g., text, images, and/or video). A user device **106** typically includes a user application, such as a web browser, to facilitate the sending and receiving of data over the network **102**, but native applications executed by the user

device **106** can also facilitate the sending and receiving of data over the network **102**.

(11) User device **106** includes software **107**. Software **107** can be, for example, a browser or an operating system. In some implementations, software **107** allows a user to access information through a network, such as network **102**, retrieving information from a server and displaying the information on a display of user device **106**. In some implementations, software **107** manages user device **106**'s hardware and software resources and provides common services for other programs on user device **106**. Software **107** can act as an intermediary between programs and user device **106**'s hardware.

(12) Software **107** is specific to each user device **106**. As described in detail below, the privacy preserving data analysis and collection innovations provide a device-specific solution that is resource-efficient and secure.

(13) An electronic document is data that presents a set of content at a user device **106**. Examples of electronic documents include webpages, word processing documents, portable document format (PDF) documents, images, videos, search results pages, and feed sources. Native applications (e.g., “apps”), such as applications installed on mobile, tablet, or desktop computing devices are also examples of electronic documents. Electronic documents **105** (“Electronic Docs”) can be provided to user devices **106** by electronic doc servers **104**. For example, the electronic doc servers **104** can include servers that host publisher websites, such as a network domain (e.g., eTLD+1). The electronic doc servers **104** can each be servers within or associated with a separate domain (e.g., a different eTLD+1).

(14) In this example, the user device **106** can initiate a request for a given publisher webpage, and the electronic document server **104** that hosts the given publisher webpage can respond to the request by sending machine Hyper-Text Markup Language (HTML) code that initiates presentation of the given webpage at the user device **106**.

(15) Electronic documents can include a variety of content. For example, an electronic document **105** can include static content (e.g., text or other specified content) that is within the electronic document itself and/or does not change over time. Electronic documents can also include dynamic content that may change over time or on a per-request basis. For example, a publisher of a given electronic document can maintain a data source that is used to populate portions of the electronic document. In this example, the given electronic document can include a tag or script that causes the user device **106** to request content from the data source when the given electronic document is processed (e.g., rendered or executed) by a user device **106**. The user device **106** integrates the content obtained from the data source into a presentation of the given electronic document to create a composite electronic document including the content obtained from the data source.

(16) In some situations, a given electronic document can include a digital content tag or digital content script that references the DCDS **110**. In these situations, the digital content tag or digital content script is executed by the user device **106** when the given electronic document is processed by the user device **106**. Execution of the digital content tag or digital content script configures the user device **106** to generate a request **108** for digital content, which is transmitted over the network **102** to the DCDS **110**. For example, the digital content tag or digital content script can enable the user device **106** to generate packetized data request including a header and payload data. The request **108** can include data such as a name (or network location) of a server from which the digital content is being requested, a name (or network location) of the requesting device (e.g., the user device **106**), and/or information that the DCDS **110** can use to select digital content provided in response to the request. The request **108** is transmitted, by the user device **106**, over the network **102** (e.g., a telecommunications network) to a server of the DCDS **110**.

(17) The request **108** can include data that specifies the electronic document and characteristics of locations at which digital content can be presented. For example, data that specifies a reference (e.g., URL) to an electronic document (e.g., webpage) in which the digital content will be presented, available locations (e.g., digital content slots) of the electronic documents that are

available to present digital content, sizes of the available locations, positions of the available locations within a presentation of the electronic document, and/or media types that are eligible for presentation in the locations can be provided to the DCDS **110**. Similarly, data that specifies keywords designated for the selection of the electronic document (“document keywords”) or entities (e.g., people, places, or things) that are referenced by the electronic document can also be included in the request **108** (e.g., as payload data) and provided to the DCDS **110** to facilitate identification of digital content items, such as electronic docs or digital components, that are eligible for presentation with the electronic document.

(18) Requests **108** can also include data related to other information, such as information that the user has provided, geographic information that indicates a state or region from which the request was submitted, or other information that provides context for the environment in which the digital content will be displayed (e.g., a type of device at which the digital content will be displayed, such as a mobile device or tablet device). User-provided information can include demographic data for a user of the user device **106**. For example, demographic information can include age, gender, geographical location, education level, marital status, household income, occupation, hobbies, social media data, and whether the user owns a particular item, among other characteristics.

(19) Data that specifies characteristics of the user device **106** can also be provided in the request **108**, such as information that identifies a model of the user device **106**, a configuration of the user device **106**, or a size (e.g., physical size or resolution) of an electronic display (e.g., touchscreen or desktop monitor) on which the electronic document is presented. Requests **108** can be transmitted, for example, over a packetized network, and the requests **108** themselves can be formatted as packetized data having a header and payload data. The header can specify a destination of the packet and the payload data can include any of the information discussed above.

(20) Further to the privacy preserving techniques discussed throughout this document, a user may be provided with controls allowing the user to make an election as to both if and when systems, programs, or features described herein may enable collection of user information (e.g., information about a user's social network, social actions, or activities, profession, a user's preferences, or a user's current location), and if the user is sent content or communications from a server. In addition, certain data may be treated in one or more ways before it is stored or used, so that personally identifiable information is removed. For example, a user's identity may be treated so that no personally identifiable information can be determined for the user, or a user's geographic location may be generalized where location information is obtained (such as to a city, ZIP code, or state level), so that a particular location of a user cannot be determined. Thus, the user may have control over what information is collected about the user, how that information is used, and what information is provided to the user.

(21) The DCDS **110** selects digital content that will be presented with the given electronic document in response to receiving the request **108** and/or using information included in the request **108**. In some implementations, the DCDS **110** is implemented in a distributed computing system (or environment) that includes, for example, a server and a set of multiple computing devices that are interconnected and identify and distribute digital content in response to requests **108**. The set of multiple computing devices operate together to identify a set of digital content that is eligible to be presented in the electronic document from among a corpus of millions or more of available digital content. The millions or more of available digital content can be indexed, for example, in a digital component database **112**. Each digital content index entry can reference the corresponding digital content and/or include distribution parameters (e.g., selection criteria) that condition the distribution of the corresponding digital content.

(22) The identification of the eligible digital content can be segmented into multiple tasks that are then assigned among computing devices within the set of multiple computing devices. For example, different computing devices can each analyze a different portion of the digital component database **112** to identify various digital content having distribution parameters that match

information included in the request **108**.

(23) The DCDS **110** aggregates the results received from the set of multiple computing devices and uses information associated with the aggregated results to select one or more instances of digital content that will be provided in response to the request **108**. In turn, the DCDS **110** can generate and transmit, over the network **102**, reply data **114** (e.g., digital data representing a reply) that enables the user device **106** to integrate the select set of digital content into the given electronic document, such that the selected set of digital content and the content of the electronic document are presented together at a display of the user device **106**.

(24) The DCDS **110** can forward requests **108** from software **107** of a user device **106** to the sources of the data, such as the electronic doc servers **104** and can forward the replies **114** from the electronic doc servers **104** to the software **107** of the user device **106**. For example, the DCDS **110** acts as a middle-man between the electronic doc servers **104** and the user devices **106** and/or the software **107** running on the user devices **106**.

(25) Randomized cohort generator **121** (RCX Generator) allows an electronic doc server **104** to generate a randomized cohort, the privacy-preserving monitoring/aggregation mechanism described herein. Within this document, a randomized cohort refers to a specific format of the privacy-preserving monitoring mechanism having a randomized cohort identifier and a randomized cohort timestamp. A randomized cohort is generated in response to an initial third party request from an application such as software **107** or a device such as user device **106**, and includes an identifier (i.e., a randomized cohort identifier) and a timestamp (i.e., a randomized cohort timestamp). For example, the randomized cohort can be data represented by rcx (rcx.id, rcx.timestamp), where rcx represents the randomized cohort, rcx.id represents the randomized cohort identifier and rcx.timestamp represents the timestamp. The randomized cohort is assigned to the software **107** or user device **106** from which the initial request was received. For simplicity of explanation, in this example, randomized cohorts are generated in response to an initial request from a browser **107**. In other examples, randomized cohorts can be generated in response to an initial request from a particular user device **106**.

(26) Because randomized cohort generator **121** is associated with a particular domain, the randomized cohort generated by each generator **121** can be domain-scoped, meaning the randomized cohort data is used within the domain with which the randomized cohort generator **121** and/or the electronic doc server **104** is associated, and that the randomized cohort is not provided to, or shared with, other domains or servers.

(27) The initial request can be a request for an electronic doc **105** from electronic doc server **104**. The initial request can be a request for a content item from a third party server **150** that provides content such as digital components that can be provided for display along with content requested from the electronic doc servers **104**.

(28) The randomized cohort identifier is a randomly selected or constructed identifier that is also assigned to a number of other browsers **107**, executing on other user devices **106**, that have also provided initial requests to the electronic doc server **104**. For example, the randomized cohort identifier can be a randomly generated 64 bit identifier that is selected from among a set of existing identifiers or created in response to the initial request from the browser **107**. The number of other applications **107** to which the randomized cohort identifier is assigned is based on a predetermined threshold level of privacy guaranteed to users. For example, the electronic doc servers **104** can implement a guarantee of k-anonymity, meaning each randomized cohort identifier is assigned to at least k browsers **107** operating on different user devices **106**. Each electronic doc server **104** can independently select a k to guarantee. In some examples, each electronic doc server **104** guarantees the same level of k-anonymity.

(29) The randomized cohort timestamp indicates a time at which the randomized cohort identifier is requested and/or assigned to the software **107** in response to the request **108**. For example, the randomized cohort timestamp can indicate the time at which the request **108** was received by the

randomized cohort generator **121**. In another example, the randomized cohort timestamp can indicate the time at which the randomized cohort identifier was selected in response to the request **108**. In another example, the randomized cohort timestamp can indicate the time at which the randomized cohort identifier was assigned to software **107** in response to request **108**. One or more of these actions can be performed at the same time, and therefore the randomized cohort timestamp can represent the time at which one or more of these actions is performed. Because the randomized cohort identifier is assigned to at least k browsers **107** for the purposes of maintaining k -anonymity (i.e., assigned to 3000 different browsers **107** for $k=3000$), the combination of the randomized cohort timestamp and the randomized cohort identifier can act as a unique identifier. In order to preserve privacy when providing the randomized cohort, randomized cohort generator **121** can anonymize the randomized cohort timestamp as well, creating a parameter representing an age bucket within which the randomized cohort falls. An age bucket represents a generalized range of age values within which the age of the randomized cohort falls, but cannot be used to uniquely identify the browser **107** to which the randomized cohort was assigned. For example, randomized cohort generator **121** can determine a difference between a current time and the randomized cohort timestamp to determine an age of the randomized cohort. Randomized cohort generator **121** can then generate a value for the age bucket based on, for example, information such as the k value and a range of ages needed to maintain k -anonymity or a predetermined range of ages, among other parameters.

(30) In addition to the randomized cohort, which includes the randomized cohort identifier and the randomized cohort timestamp, randomized cohort generator **121** generates a certificate that can be used to attest to the validity of the randomized cohort. For example, randomized cohort generator **121** can generate a certificate that contains a public verification key that is signed by the electronic doc server **104**. The electronic doc server **104** can also generate a public/private key pair. The certificate generation process and verification process is described in further detail below.

(31) The randomized cohort, which includes both of the randomized cohort identifier and the randomized cohort timestamp, is a flexible privacy-preserving monitoring mechanism that allows for anonymity as well as unique identification. As described in further detail below, the certificate can be used to uniquely identify the browser **107** when the browser **107** is attesting to the validity of its randomized cohort. For example, the browser **107** can transmit, for verification purposes, the certificate provided by randomized cohort generator **121** to a verification system.

(32) In some implementations, no metadata other than the randomized cohort is provided by the issuing domain, or electronic doc server **104**, to be stored on the user device **106** on which the browser **107** is stored. This additional restriction further improves user privacy by reducing the amount of data that is collected and stored, eliminating the possibility of compromising particular types of user data, which is not collected and therefore cannot be linked to a particular user or randomized cohort identifier.

(33) Analyzer **123** analyzes randomized cohort data to monitor user web activity. Analyzer **123** can receive a randomized cohort identifier and randomized cohort age data with requests for data from the electronic doc server **104** with which analyzer **123** is associated, and use the received randomized cohort identifiers and randomized cohort age data to perform security functions. For example, analyzer **123** can detect, based on randomized cohort identifiers and randomized cohort age data, certain types of fraudulent activity or coordinated abuse of the system or content from electronic doc server **104**. As illustrated in FIG. 1, each electronic doc server **104** can have a separate analyzer **123** tailored to its own needs. In some examples, electronic doc servers **104** can share a centralized analyzer **123**, which can be implemented as a remote or separate analysis server or service.

(34) System **100** includes one or more third party independent verification services that a user of a user device **106** or browser **107** can choose to use. The third party independent verification services independently verify privacy properties of randomized cohorts assigned by the issuing services,

such as the electronic doc servers **104** as described above and the trusted domain server **140** as described below. The independent verification of the privacy properties of the randomized cohorts is optional for the user and is described in further detail below.

(35) Verification server **130** is a server, independent of the electronic doc servers **104**, that performs verification of the statistical properties of randomized cohort identifiers and pairs of randomized cohort identifiers and randomized cohort age parameters. Verification server **130** acts as an independent server that does not issue randomized cohorts and does not participate in monitoring randomized cohort data or otherwise interacting with servers, such as electronic doc servers **104**, that monitor and/or analyze randomized cohort data. Verification server **130** allows users to verify that issuing servers, such as electronic doc servers **104**, are maintaining the guaranteed level of privacy for the particular randomized cohort. By giving users a chance to verify, through an independent service, that their privacy is being maintained by participating issuing domains, the system **100** encourages user trust and improves user experience. Additionally, this allows users to discern whether a particular issuing domain is in compliance and hold issuing domains accountable, thus improving the experience for all users.

(36) Trusted domain server **140** is a server, independent of the electronic doc servers **104**, that can issue a randomized cohort to software **107** in response to a request **108**. Trusted domain server **140** issues randomized cohorts that are globally-scoped, meaning the randomized cohorts can be provided to requesting servers from different domains, and is not limited to use within a particular domain like the domain-scoped randomized cohorts generated by the electronic doc servers **104**. Trusted domain server **140** acts as a central source of randomized cohorts, guaranteeing one or more levels of privacy for each randomized cohort generated and assigned. In some implementations, trusted domain server **140** is separate from the electronic doc servers **104**, does not share information with any of the electronic doc servers **104**, and does not provide or host content. For example, trusted domain server **140** does not participate in the content distribution process, and is involved in the system **100** only to generate and assign randomized cohorts to maintain the privacy of users of system **100**.

(37) Randomized cohort generator **142** (RCX Generator) is a generator that operates similarly to the randomized cohort generator **121** as described above, but is associated with the trusted domain server **140** instead of an electronic doc server.

(38) FIG. 2 is a data flow diagram of an example process **200** for issuing and implementing a privacy-preserving monitoring mechanism. As described below, domains supporting randomized cohorts are required to assign randomized cohorts to users in a way that preserves certain verifiable k-anonymity properties. Operations of the process **200** can be implemented, for example, by electronic doc servers **104**, user devices **106**, verification server **130**, trusted domain server **140**, and/or third party servers **150**. Operations of the process **200** can also be implemented as instructions stored on one or more computer readable media which may be non-transitory, and execution of the instructions by one or more data processing apparatus can cause the one or more data processing apparatus to perform the operations of the process **200**.

(39) As discussed above with respect to FIG. 1, the DCDS **110** may act as an intermediary, forwarding requests **108** from user devices **106** to electronic doc servers **104** and forwarding replies **114** from electronic doc servers **104** to user devices **106**. In this particular example, the DCDS **110** may transmit data between user devices **106** and electronic doc servers **104**, but is not illustrated in the data flow.

(40) The process **200** begins with stage A-1, in which user device **106** provides a request to electronic doc servers **104** for content. For example, software **107** provides a request to an electronic doc server **104**, associated with a specific network domain, for content such as a particular web page. Software **107** can provide a request for third party content, such as digital component, in addition to the web page. For example, software **107** can provide a request to third party servers **140**. The request in stage A-1 can be the request from the user device **106** that is

either a request for the content or a digital component. The request can be provided to an electronic doc server **104** or to a third party server **140**. In this particular example, the request, labelled request **1**, is provided to an electronic doc server **104**.

(41) Request **1** is an initial request from software **107**. An initial request indicates that software **107** has not previously issued a request to electronic doc server **104**, that software **107** has not previously been issued a randomized cohort, and/or that a previously issued randomized cohort has expired, been reset by the user associated with the software **107**, or is otherwise invalid. For example, request **1** can indicate that software **107** has not previously issued a request to electronic doc server **104**, and thus that electronic doc server **104** should issue and assign a randomized cohort to software **107**. In some implementations, if system **100** uses a single, central issuing entity such as trusted domain server **140**, stage A-2 occurs, in which data indicating request **1** is forwarded from electronic doc servers **104** to trusted domain server **140**. If system **100** uses separate issuing entities, such as electronic doc server **104**, stage A-2 does not occur. In this particular example, stage A-2 does not occur because system **100** uses separate issuing entities. In other examples and implementations, stage A-2 can occur even if system **100** uses separate issuing entities.

(42) The process **200** continues in stage B, in which a randomized cohort generator generates/constructs a randomized cohort **201** in response to receiving data indicating request **1** and assigns the randomized cohort to the requesting user device **106** or software **107**.

(43) If system **100** uses a single, central issuing entity such as trusted domain server **140**, randomized cohort generator **142** performs stage B by generating a randomized cohort in response to receiving data indicating request **1**. If system **100** uses separate issuing entities, such as electronic doc server **104**, randomized cohort generator **121** performs stage B by generating a randomized cohort in response to receiving data indicating request **1**. In this particular example, stage B is described as being performed by randomized cohort generator **121**. In other examples and implementations, stage B can be performed by randomized cohort generator **142** if, for example, system **100** uses a single, central issuing entity such as trusted domain server **140**.

(44) Randomized cohort generator **121** generates each randomized cohort based on a randomly selected identifier and a timestamp. An identifier can be randomly generated or, when system **100** has a sufficient volume of different browsers **107** running on different user devices **106** or different user devices **106**, an identifier can be selected from among existing identifiers that are already in use. In this particular example, the randomized cohort generator **121** selects a 64-bit identifier as the randomized cohort identifier.

(45) The concept of k-anonymity as applied to randomized cohort identifiers guarantees that each randomized cohort identifier is assigned to at least k browsers **107**. For example, the k different browsers **107** can be used as a proxy metric to guarantee k different user devices **106** corresponding to k different users. Each of the issuing domains guarantees a particular k for the randomized cohorts that it issues, such that each of the randomized cohort identifiers is assigned to at least k browsers **107** running on different user devices **106**. In order to guarantee such levels of privacy, the issuing domains must track the number of each randomized cohort identifier that is issued and associate browsers **107** with particular randomized cohort identifiers that have already been assigned. During the assignment process browsers **107** can be assigned to clusters of browsers **107** assigned to a particular randomized cohort identifier. For example, randomized cohort generator **121** can randomly assign browser **107**, from which request **1** was received, to a cluster of browsers **107** that are associated with a particular randomized cohort identifier.

(46) Each issuing domain performs the assignment process to provide a uniformly distributed (or within a threshold of uniformly distributed) assignment of identifiers. For example, randomized cohort generator **121** can randomly assign browser **107** to one of a set of randomized cohort identifiers that have already been generated and associated with browsers **107**. Randomized cohort generator **121** balances the number of different randomized cohort identifiers with the number of browsers in order to maintain its k-anonymity guarantee to users. For example, randomized cohort

generator **121** can balance the number of different randomized cohort identifiers being assigned based on the number of expected and current browsers **107** providing requests for content to the respective electronic doc servers **104**. In one example, randomized cohort generator **121** can generate or receive a threshold number of different randomized cohort identifiers based on expected browsers **107** providing requests for content to the electronic doc server **104** and can adjust the threshold number of different randomized cohort identifiers based on an actual, current amount of traffic from different browsers **107** providing requests for content to the electronic doc server **104**. For example, randomized cohort generator **121** can increase the number of different randomized cohort identifiers associated with its domain based on an increase in the number or a velocity of change in the number of different browsers **107**. In another example, in an initialization of the system **100**, the randomized cohort generator **121** can receive a number of requests from different browsers **107** running on separate user devices **106** and randomly assign a cluster of k browsers **107** to a particular randomized cohort identifier to guarantee k -anonymity.

(47) In order to guarantee k -anonymity, a domain must have a sufficient volume of traffic for statistical arguments to become meaningful and provide an anonymizing effect such that a particular randomized cohort identifier is assigned to a number k of different browsers running on different user devices. System **100** can ensure that the system has sufficient volume of traffic by setting and adjusting thresholds for web activity levels and to ensure that a domain sustains a sufficient amount of traffic for randomized cohort privacy properties to become verifiable.

(48) Once a randomized cohort is generated and assigned to a browser **107**, the randomized cohort can have an expiration time. For example, the randomized cohort can expire after a predetermined period of time or until a user resets, or clears, the randomized cohort. In some examples, the randomized cohort can expire after a period of time specified by a user of browser **107**, a default setting of browser **107**, the issuing domain, a period of time determined by browser **107** based on the web activity of the user of browser **107** and the habits of the user or users having similar browsing habits as the user. For example, if users often clear randomized cohorts associated with an issuing domain on a regular schedule, the expiration period of a randomized cohort can be adjusted such that the randomized cohort is cleared on a regular schedule similar to the one that users of the domain generally use.

(49) Randomized cohort generator **121** can reassign previously-used randomized cohort identifiers when a randomized cohort identifier assignment expires or is cleared. For example, randomized cohort generator **121** can randomly assign a randomized cohort identifier to a browser **107** using a list of available randomized cohort identifiers, including previously expired randomized cohort identifiers, or randomly generate a randomized cohort identifier based on the needs of the system.

(50) In some implementations, randomized cohort identifiers can get “stale;” in other words, the number of users assigned to a particular randomized cohort identifier may degrade over time as users reset their randomized cohorts or their randomized cohorts expire. While an issuing domain, such as electronic doc server **104**, can guarantee k users at the time that the randomized cohorts are issued, the number of users behind an FC may decrease as randomized cohorts expire or are reset. The system **100** can counteract this reduction in users per bucket by, for example, storing one or more additional states on the server that indicate a recycling status for randomized cohort identifier/age buckets for which there are less than k users. Additionally, verification servers **130** can also detect a lack of activity in a particular bucket of users (i.e., a bucket of users assigned to a particular randomized cohort identifier within a period of time). Verification servers **130** can also provide feedback to browsers **107** regarding the distribution of randomized cohort identifiers and can provide feedback to randomized cohort generator **121**.

(51) In addition, the randomized cohort generator **121** generates a timestamp that indicates the time at which the randomized cohort identifier was assigned or generated as described above with respect to FIG. 1. For example, at the time that the request from software **107** is received, the randomized cohort generator generates a randomized cohort timestamp. The timestamp indicates

when the randomly selected identifier was assigned to a issuing domain that assigns randomized cohorts to users randomly.

(52) Users of browsers **107** can reset the randomized cohort associated with their browser **107** at any time just as other monitoring mechanisms can be clear and/or reset. When a user resets the randomized cohort issued by an issuing domain associated with their browser **107**, the next request that is sent from the browser **107** to the issuing domain, such as an electronic doc server **104** is received as an initial request for which a randomized cohort is to be assigned to the browser **104**.

(53) In addition to the randomized cohort assigned to browser **107**, randomized cohort generator **121** generates a certificate that contains the randomized cohort and a public/private key pair. This certificate, provided by randomized cohort generator **121**, contains identifiable information of browser **107**, including the randomized cohort identifier and the randomized cohort timestamp. Randomized cohort generator **121** signs the certificate, which indicates the authenticity of the certificate and its assignment to browser **107**. The signed certificate is used only for verification purposes, as described in further detail below, and is never transmitted to or accessible by domains other than a verification server. Randomized cohort generator **121** generates a public key, which may be known to others, and a private key, which may never be known by anyone except the browser **107**, using cryptographic algorithms. The public/private key pair is used for attestation, or proving the identity of the browser **107** providing the signed certificate that includes the randomized cohort. In particular, the public key is provided to a requesting entity and browser **107** can use the private key to cryptographically prove that it was assigned the certificate. In addition, the certificate can be encrypted using the public/private key pair.

(54) The verification server **130** verifies that the signature on the certificate is valid. For example, if examplecoolvideoplatform.com issued the certificate, then the verification server **130** would retrieve examplecoolvideoplatform.com's public keys and verify that the signature on the certificate is valid. The keys can, for example, be distributed through techniques such as public key infrastructure (PKI), and do not need to be distributed by the electronic doc server **104** associated with the randomized cohort generator **121** that generated the certificate. The verification server **130** then generates a challenge and transmits the challenge to the browser **107**. For example, the challenge can be a random number or some variable known to the verification server **130**. The browser **107** can then sign the challenge using its private key and return the signature to the verification server **130**. The verification server **130** is then able to verify that the signature is valid using the public verification key published on the signed certificate.

(55) The process **200** continues with stage C, in which electronic doc server **104** provides the randomized cohort, the signed certificate, the public/private key pair, and the response to the request to the browser **107**. In some implementations, electronic doc server **104** provides each of the randomized cohort, the signed certificate, the public/private key pair, and the response to the request at the same time to the browser **107**. In other implementations, electronic doc server **104** provides one or more of the randomized cohort, the signed certificate, the public/private key pair, and the response to the request separately to the browser **107**. In some implementations, the private/public key pair is distributed through standard techniques such as PKI. As described above, particular stages of process **200** that are performed by electronic doc server **104**. For example, stage C can be performed by trusted domain server **140**.

(56) The process **200** continues with stage D, in which browser **107** provides a subsequent request to electronic doc server **104**, along with randomized cohort data. The subsequent request can be identical in format to the initial request, and is not required to provide any additional information indicating that the request is subsequent to the initial request. Browser **107** detects that the request is being provided to a domain associated with a randomized cohort that browser **107** has been assigned and provides randomized cohort data, thus indicating to electronic doc server **104** that the request is a subsequent request. The randomized cohort data allows the domain associated with electronic doc server **104** to monitor the activity of different browsers within the domain while

preserving the privacy of the users of the browsers to a greater extent than was previously possible. (57) Browser **107** provides randomized cohort data including its assigned randomized cohort identifier for the domain associated with electronic doc server **104** and data representing an obscured age of the randomized cohort for the domain. Specifically, browser **107** generates a randomized cohort age bucket to be provided instead of the randomized cohort timestamp to obscure the specific identity of the browser **107**. For example, browser **107** uses a function $RCX.AgeBucketFn()$ to compute a randomized cohort age bucket. The function can be, for example, a bucketing function such as $\log_2(\text{currentTime} - \text{randomized cohort. Timestamp})$, where currentTime represents the current time and $\text{randomized cohort. Timestamp}$ represents the randomized cohort timestamp.

(58) Because browser **107** provides both of the randomized cohort identifier and the randomized cohort timestamp with subsequent requests, the issuing domain (in this example, electronic doc server **104** of the issuing domain) is responsible for guaranteeing that there are at least k users, or user proxies of browsers **107**, assigned to each pair of randomized cohort identifier and randomized cohort age bucket in order to satisfy its k -anonymity guarantee. In some implementations, the issuing domain can guarantee different levels of k -anonymity with respect to the randomized cohort identifier and the randomized cohort identifier and randomized cohort timestamp pair.

(59) Optionally, users of browser **107** can choose to verify the statistical and/or privacy properties of the randomized cohort assigned to browser **107**. Any third party that is not the issuing domain of the randomized cohort to be verified (such as electronic doc server **104** or trusted domain server **140**) can maintain a verification server **130**, and users can choose to direct their browsers **107** to any verification server **130**. In some implementations, electronic doc servers **104** can maintain a verification server **130** to verify the statistical and/or privacy properties of randomized cohorts that are issued by other electronic doc servers **104** associated with other domains. In some implementations, browsers **107** can automatically request verification from verification servers **130** without being instructed by a user. For example, system **100** can require participating browsers **107** to periodically request verification from a randomly selected eligible verification server **130**. In some implementations, browsers **107** do not request verification unless instructed by a user.

(60) The process **200** continues with stage E, in which software **107** provides the certificate to verification server **130**. For example, browser **107** provides the certificate generated by randomized cohort generator **121** in stage B to verification server **130**. Browser **107** can also provide the public key to the verification server for attestation purposes. In some implementations, browser **107** provides randomized cohort information, such as the randomized cohort identifier and the randomized cohort age bucket, instead of the certificate, which does not obscure the randomized cohort age.

(61) The process **200** continues with stage F, in which the verification server **130** performs the verification process of verifying the statistical and/or privacy properties of the randomized cohort data provided by browser **107**.

(62) The verification server **130** verifies that the set of randomized cohort identifiers to which the verification server **130** has access are uniformly distributed. For example, verification server **130** can use the certificate to determine a randomized cohort identifier and/or a randomized cohort age bucket. Verification server **130** can then compare the randomized cohort identifier and/or the randomized cohort age bucket to a list of randomized cohort identifiers and/or randomized cohort age buckets to which verification server has access to determine whether the number of browsers operating on different user devices assigned to each different randomized cohort identifier within an appropriate scope is uniformly distributed or within a threshold distance to being uniformly distributed. As discussed above, the scope can be within a particular domain or globally within the network of participating domains. For example, verification server **130** can maintain a list of randomized cohort identifier and randomized cohort age bucket pairs that it has received from browsers **107** participating in its verification service. Verification server **130** can then determine

whether the number of different browsers operating on different user devices assigned to each different randomized cohort identifier within a particular age bucket and/or domain is normally distributed. In some implementations, verification server **130** determines whether the number of different browsers operating on different user devices assigned to each different randomized cohort identifier within a particular domain is normally distributed without the additional requirement of being within the same age bucket.

(63) Browser **107** can automatically request verification from a verification service such as verification server **130** at a particular time interval. For example, browser **107** can request verification from verification server **130** on a weekly basis to ensure that electronic doc server **104** is in compliance with its k-anonymity guarantee responsibilities, and is delivering the level of privacy that it promises to users.

(64) Optionally, during a content delivery process in which randomized cohorts are globally-scoped and are not assigned by each domain, an electronic doc server **104** from which browser **107** requests content can include an attestation request in its response to the initial request from browser **107**.

(65) Such a randomized cohort would be provided by browser **107** to each electronic doc server **104** on each request, regardless of the domain with which the electronic doc server **104** is associated. Using globally-scoped randomized cohorts prevents malicious domains from colluding to combine randomized cohorts across domains in a way that creates more trackable entropy, reducing the risk of compromising user privacy due to collusion between domains. However, globally-scoped randomized cohorts are globally readable, which does not eliminate the possibility of abuse due to tactics such as replay attacks. For example, a malicious actor can purchase traffic to a domain, harvest randomized cohorts (including randomized cohort identifiers and randomized cohort timestamps) and then use the observed distributions to attack a different domain. Because the entity that produced the randomized cohorts is separate and independent different from the servers from which content is requested, such as electronic doc servers **104** or third party servers **150**, the server from which content is requested would need to be able to perform an attestation step to verify that the browser **107** providing the randomized cohort identifier and randomized cohort timestamp has actually been assigned a randomized cohort from a trusted server such as trusted domain server **140**.

(66) The attestation request can specify browser **107** should request for a verification server **130** to determine whether the randomized cohort identifier and randomized cohort age bucket provided to the electronic doc server **104** with a request for content by browser **107** was actually assigned to browser **107** by a trusted server, such as trusted domain server **140**. For example, electronic doc server **104** can request that verification server **130** require browser **107** to attest to its identity as the browser **107** to which the certificate and associated randomized cohort information was assigned by a trusted server, such as trusted domain server **140**. Verification server **130** can facilitate this attestation process by generating an anonymized certificate, different from the certificate provided and signed by randomized cohort generator **121**, that attests that a particular certificate was assigned to a particular browser **107**. This process is not relevant where the randomized cohorts are locally-scoped to each domain such that each domain issues a randomized cohort to the browser **104**, and therefore would be able to determine whether browser **107** is the browser to which it assigned a randomized cohort based on the signed certificate issued by randomized cohort generator **121**. For example, if the randomized cohort generator **121** associated with electronic doc server **104** generated the randomized cohort, it would be able to verify its own signature on a certificate that it receives associated with the randomized cohort or through the use of public key encryption using the public/private key pair generated as described above with respect to stage B.

(67) In one illustrative example, a user accesses ExampleNewsWebsite.com using browser **107**. The user reads an article that links to a particular video on ExampleVideoHostingPlatform.com. In this example, browser **107** has previously been assigned a randomized cohort and certificate from

trusted domain server **140**, and so browser **107** sends a request for the particular video to ExampleVideoHostingPlatform.com, which is associated with electronic doc server **104**. In the request, browser **107** includes its assigned randomized cohort identifier and randomized cohort age bucket.

(68) Because the globally-scoped randomized cohort assigned to browser **107** by a trusted server was not generated by randomized cohort generator **121** associated with electronic doc server **104**, ExampleVideoHostingPlatform.com may request that browser **107** attest to its identity and forward this request to verification server **130** to complete. Electronic doc server **104** provides browser **107** with a secret X. The secret can have any value, and can be, for example, a randomly generated 16 bit value. Browser **107** provides the secret X and the certificate provided by randomized cohort generator **142** to verification server **130** to request that verification server **130** perform the attestation process. Browser **107** can blind the value of X from verification server **130** to prevent the secret X from being linked to browser **107**. For example, browser **107** can use a partially blind signature scheme to blind X from verification server **130**. Verification server **130** then receives browser **107**'s certificate and the blinded secret X and generates an anonymized certificate that indicates the randomized cohort identifier and a randomized cohort age bucket based on the certificate. Verification server **130** then signs this anonymized certificate and returns the signed, anonymized certificate and X to electronic doc server **104** that issued the attestation request. Electronic doc server **104** can compare the randomized cohort information in the signed, anonymized certificate from the verification server **130** to the randomized cohort identifier and randomized cohort age bucket that electronic doc server **104** received from browser **107**. If the information matches, browser **107** has successfully attested to its identity has the browser to which trusted domain server **140** assigned the randomized cohort information that was provided to electronic doc server **104**.

(69) In some implementations, a more robust verification process can be performed if multiple verification servers **130** collaborate and/or share resources. For example, verification servers **130** can determine whether a number of randomized cohort identifiers within a particular age bucket is in compliance with a k-anonymity guarantee provided by the electronic doc server **104**, specified by system **100** in order to be considered in compliance, etc. in the context of a larger population that better represents the total population of participating browsers and domains. Verification servers **130** can use, for example, the total number of unique public keys provided with certificates received across every verification servers can be computed, allowing a user to verify that there are approximately at least k users that have the same randomized cohort identifier.

(70) In another example implementation, electronic doc server **104** can return a unique tracking URL to browser **107** instead of a secret X. Browser **107** can then follow the unique tracking URL and post the signed anonymized certificate from the verification server **130** to the destination at the unique tracking URL.

(71) The independent verification servers **130** can collaborate to show that certain statistical properties of an entire population of randomized cohorts are satisfied and that privacy guarantees are being upheld at scale. Due to the nature and scale of the number of browsers running on different user devices representing different users, the verification servers **130** provide a level of protection in the form of check points that are difficult to spoof, and not formal proofs of correctness or legitimacy.

(72) In some examples, a malicious issuer can use a subset of bits of the randomized cohort identifier to encode sensitive data about a user or otherwise embed information in a hidden way. System **100** can perform uniformity checks at the bit level to inspect the randomized cohort identifiers and determine whether the randomized cohort identifiers are uniformly selected and distributed. For example, verification server **130** can randomly generate a mask, perform a bitwise AND operation with all randomized cohort identifiers and re-aggregate. If randomized cohort identifiers have been uniformly selected and assigned, the result should also be uniform for every

mask. In some implementations, verification server **130** remembers randomized cohort identifiers that have been previously assigned for each user and test for bit-level correlations over time. In some implementations, verification server **130** performs tests to ensure that two or more browsers **107** running on different user devices **106** are not consistently assigned to the same randomized cohort identifier.

(73) The process **200** continues with stage G, in which verification server **130** provides the results of the verification process to browser **107**. Verification server **130** can provide the statistical and/or privacy properties of the randomized cohort identifier and randomized cohort age bucket to browser **107** in terms of unprocessed numbers. For example, verification server **130** can provide the distribution of randomized cohort identifier assignments or the number of different randomized cohort browsers running on different user devices to browser **107** associated with the randomized cohort identifier. Browser **107** can then compare the unprocessed numbers distribution of randomized cohort identifier assignments to a threshold deviation from a uniform distribution or the number of different randomized cohort browsers running on different user devices to browser **107** associated with the randomized cohort identifier to a threshold number k of different browsers running on different user devices that should be associated with the randomized cohort identifier in order to guarantee k -anonymity. Browser **107** can provide an indication to a user of browser **107** when one or more thresholds have not been satisfied. For example, if the number of different randomized cohort browsers running on different user devices to browser **107** associated with the randomized cohort identifier did not meet the threshold number k of different browsers running on different user devices, browser **107** can display a visual message, play audio, create a vibration, etc. to indicate to the user that the threshold was not met.

(74) In addition to the advantages of providing a monitoring mechanism that provides an independently verifiable level of privacy for users, the randomized cohort provides protections for content providers and hosts of the system. One vector of abuse in content distribution systems is through manipulation of engagement statistics. For example, in schemes in which a content provider pays per interaction (e.g., a pay-per-click system) a user can be incentivized to click on a particular content item to increase the payment required from the content provider. Additionally, on video content platforms, users may collude to coordinate the manipulation of a video content recommendation system by artificially boosting the popularity of certain videos by providing fraudulent views. The randomized cohort provides a monitoring mechanism that allows for the detection of such abuse.

(75) From a user perspective, the randomized cohorts function similarly to traditional monitoring mechanisms while providing a higher level of privacy. Thus, the described system for privacy-preserving monitoring mechanism requires few to no changes to a user's experience while improving their privacy and reducing the probability that a user's information is compromised.

(76) The randomized cohorts rely on the use of statistical arguments for guaranteeing user privacy, and exploit the use of much larger amounts of entropy for purposes of performing abuse detection without violating user privacy. For $k=1$ in the k -anonymity context, randomized cohorts provide the same level of abuse detection as traditional tracking mechanisms such as cookies. As k increases, randomized cohorts provide users with increasing levels of privacy while remaining useful for abuse detection. Additionally, randomized cohorts do not require trust to be asserted in a first-party context, in which the randomized cohorts are locally-scoped and limited to use by the domain that issued the randomized cohort.

(77) Once browser **107** has provided randomized cohort information to the electronic doc server **104**, the electronic doc server **104** can perform security functions such as statistical analysis to detect abuse of the system **100**. For example, when the randomized cohorts are locally-scoped to a particular domain and when the randomized cohorts are globally-scoped, each analyzer **123** of a particular electronic doc server **104** can perform statistical analysis as specified by the electronic doc server **104** with which the analyzer **123** is associated.

- (78) An electronic doc server **104** can protect its domain against engagement abuse in a way that preserves user privacy through the use of the described monitoring mechanism, randomized cohorts. Because the system guarantees k-anonymity where k is adjustable, the use of randomized cohorts provides a system that offers a continuum of protection from engagement abuse that maintains a level of privacy protection higher than is current possible.
- (79) Analyzer **123** of electronic doc server **104** monitors anonymized user web activity through the use of the randomized cohort information and detects signs of engagement abuse or collusion based on statistical properties of the distribution of randomized cohort identifiers and the activity associated with the randomized cohort identifiers. For example, analyzer **123** can detect statistical outlier behavior, such as irregular resource requests or unusual activity from a group of users associated with a particular randomized cohort identifier and/or age bucket.
- (80) For a given participating domain, the number of clicks per randomized cohort identifier should be uniformly distributed. A lack of uniformity in the number of clicks can indicate unusual activity. For example, a deviation greater than a threshold deviation from the number of clicks associated with a randomized cohort identifier, or a spike of activity over a period of time (the amount or velocity of deviation) associated with one or more randomized cohorts can indicate statistically unusual activity that can indicate abuse. Analyzer **123** can determine, for example, statistical tests to verify the statistically anomalous activity through metrics such as risk ratios for the activity and its effect on domain resources as compared to the risk and potential damage caused by malicious activity.
- (81) Analyzer **123** can compute age distributions for the randomized cohort age buckets received in combination with the randomized cohort identifiers and test for anomalies against, for example, a known global background distribution. For example, analyzer **123** can detect that an amount of activity from browsers with randomized cohort age bucket values below a threshold age value is at or below a threshold amount and determine that there is an unusual amount of activity.
- (82) Analyzer **123** can detect coordinated attacks involving cohorts of malicious users by performing fraud detection using algorithms that are camouflage-resistant, provides an upper bound on the effectiveness of fraud using provable limits on undetectable fraud. For example, analyzer **123** can bound the probability that two or more randomized cohort identifier/randomized cohort age bucket pairs visit the same set of long tail websites over some time window. In another example, analyzer **123** can construct a bipartite graph between users and websites and then attempting to find bicliques (a particular kind of bipartite graph where every vertex of the first set is connected to every vertex of the second set) corresponding to clusters of users acting in a coordinated way.
- (83) Analyzer **123** can detect fraudulently created entropy for the purposes of tracking one or more users. For example, a malicious actor can attempt to assign a single browser **107** to a randomized cohort identifier where the other k-1 browsers **107** are bots. However, the electronic doc server **104** assigns randomized cohort identifiers to users in a random way, and therefore attacks of this nature this attack would be difficult to pull off without an attacker obtaining server access. A malicious, or compromised domain electronic doc server **104** may be able to perpetrate this fraud on a number of users associated with browsers **107**, but it would be difficult for a malicious domain to execute an attack of any meaningful scale while evading detection by an independent verification server such as verification server **130**. In addition to providing notice of such randomized cohort identifier/age bucket pairs to electronic doc servers **104** for participating domains so that the domains can prevent abuse from the identified randomized cohort identifiers, such scaled attacks can be effectively deterred by allowing verification servers **130** to perform abuse detection and prevention, allowing verification servers **130** to detect and block or remove randomized cohorts assigned to botnets for a period of time).
- (84) When analyzer **123** detects statistically unusual behavior, analyzer **123** can set or adjust restrictions to prevent particular users from requesting resources from a particular electronic doc

server **104** or generally from requesting resources for a period of time. For example, analyzer **123** can detect that a particular randomized cohort identifier/randomized cohort age bucket pair is being provided to the electronic doc server **104** and is associated with an unusual amount of activity within the past five minutes and prevent the particular randomized cohort identifier/randomized cohort age bucket from requesting resources from electronic doc server **104**. Thus, system **100** can reduce the amount of resources used in furtherance of fraudulent activity. However, when the randomized cohorts are independent, domain-scoped randomized cohorts, each verification server **130** has access to only the randomized cohort identifiers and randomized cohort age buckets of users who have chosen to use that particular verification server **130** as their independent verification service. In some implementations, regardless of whether the randomized cohort is domain or globally-scoped, the verification servers **130** are federated and grouped with other verification servers **130** in order to have access to a threshold amount of randomized cohorts in order to provide browsers **107** with statistically meaningful results.

(85) When the randomized cohort is globally-scoped, analyzer **123** can communicate with trusted domain server **140** to identify suspected fraudulent or unusual activity associated with the globally-scoped randomized cohort identifier/randomized cohort age bucket pair, which can be communicated to other electronic doc servers **104** to monitor for and prevent abuse across participating domains.

(86) FIGS. **3A** and **3B** are swim lane diagrams that illustrate example processes **300** and **350** for issuing and implementing a privacy-preserving monitoring mechanism that is used across web activity performed across different domains. Operations of the processes **300** and **350** can be implemented, for example, by user device **106** and/or software **107**, electronic doc servers **104**, DCDS **110**, verification server **130**, trusted domain server **140**, and/or third party server **150**. Operations of the processes **300** and **350** can also be implemented as instructions stored on one or more computer readable media which may be non-transitory, and execution of the instructions by one or more data processing apparatus can cause the one or more data processing apparatus to perform the operations of the processes **300** and **350**.

(87) Referring now to FIG. **3A**, process **300** begins with step **1**, in which an initial request from software **107** is provided to electronic doc server **104**. In some examples, this request can be forwarded to electronic doc server **104** through DCDS **110**, which is not illustrated in FIG. **3A**. For example, ExampleImageHostingPlatform.com, associated with the ExampleImageHostingPlatform network domain, can receive a request from browser **107** for an image. In this example, browser **107** has not been assigned a randomized cohort from ExampleImageHostingPlatform's electronic doc server **104**, and therefore the request is an initial request. In another example, if browser **107** was previously assigned a randomized cohort from ExampleImageHostingPlatform's electronic doc server **104** but the randomized cohort has expired or been cleared by a user of browser **107**, the request is also considered an initial request.

(88) Process **300** continues with step **2**, in which electronic doc server **104**, generates a randomized cohort, including a randomized cohort identifier, a randomized cohort timestamp, a signed certificate, and a public/private key pair and assigns the randomized cohort to software **107**. For example, randomized cohort generator **121** of electronic doc server **104** can generate a randomized cohort, signed certificate, and public/private key pair and assign the randomized cohort to browser **107** as described above with respect to FIG. **2**. As described above, electronic doc server **104** is associated with a particular network domain.

(89) Process **300** continues with step **3**, in which electronic doc server **104** provides a reply to software **107** with the requested content and the randomized cohort, including the randomized cohort identifier, the randomized cohort timestamp, the signed certificate, and the public/private key pair as described above with respect to FIG. **2**. For example, electronic doc server **104** can transmit a response including the requested image, the randomized cohort, the signed certificate, and the public/private key pair to browser **107**.

(90) Process **300** continues with step **4**, in which software **107** provides a subsequent request to electronic doc server **104**, which had previously issued the randomized cohort to software **107**. Software **107** includes the randomized cohort identifier and an age bucket with its request. For example, electronic doc server **104** for domain ExampleImageHostingPlatform.com can receive a request from browser **107** for a second image as well as the assigned randomized cohort identifier and randomized cohort age bucket.

(91) Optionally, the user of software **107** can choose a verification server to independently verify the privacy and/or statistical properties of the randomized cohort identifier associated with software **107** as described with respect to steps **5**, **6**, and **7**.

(92) Process **300** continues with step **5**, in which software **107** provides its certificate to verification server **130**. For example, browser **107** can provide its signed certificate, generated by randomized cohort generator **121**, to verification server **130**.

(93) Process **300** continues with step **6**, in which verification server **130** performs the verification process based on the certificate. For example, verification server **130** can perform the verification process based on the certificate to determine statistical and/or privacy preserving properties of the randomized cohort identifier provided in the certificate.

(94) Optionally, verification server **130** can also authenticate the certificate provided prior to performing the verification process to determine that the randomized cohort identifier provided was assigned to the software **107** that is providing the certificate. For example, verification server **130** can use the public/private key pair to determine that the randomized cohort identifier provided was assigned to browser **107** that is providing the certificate.

(95) Process **300** continues with step **7**, in which verification server **130** provides the results of the verification process to software **107**. For example, verification server **130** can output unprocessed numbers to browser **107** or a result of a statistical analysis and comparison to thresholds to determine whether statistically unusual or fraudulent activity has been detected, or whether the guaranteed level of privacy was preserved for browser **107**.

(96) Referring now to FIG. **3B**, the process **350** for issuing and implementing a privacy-preserving monitoring mechanism that is used across web activity performed across different domains is performed by a central trusted domain server **140** instead of separate electronic doc servers **104**. The randomized cohorts issued by trusted domain server **140** are globally-scoped. Process **350** begins with step **1**, in which an initial request from software **107** is provided to electronic doc server **104**. In some examples, this request can be forwarded to electronic doc server **104** through DCDS **110**, which is not illustrated in FIG. **3B**.

(97) Process **350** continues with step **2**, in which electronic doc server **104** forwards the request information to trusted domain server **140**. For example, electronic doc server **104** can receive a request for several lines of text from its domain, ExampleTextandImageHostingPlatform.com. In this example, browser **107** has not been assigned a randomized cohort and does not provide one to electronic doc server **104**, and therefore the request is an initial request.

(98) Process **350** continues with step **3**, in which trusted domain server **140** generates a randomized cohort, including a randomized cohort identifier, a randomized cohort timestamp, a certificate, and a public/private key pair and assigns the randomized cohort to software **107**. For example, randomized cohort generator **142** of trusted domain server **140** can generate a randomized cohort, signed certificate, and public/private key pair and assign the randomized cohort to browser **107** as described above with respect to FIG. **2**. As described above, trusted domain server **140** is not associated with one particular network domain and generates a globally-scoped randomized cohort that can be used across different domains.

(99) Process **350** continues with step **4A**, in which electronic doc server **104** provides a reply to software **107**. For example, electronic doc server **104** can transmit a response including the requested text to browser **107**.

(100) Process **350** continues with step **4B**, in which trusted domain server **140** provides the

randomized cohort, including the randomized cohort identifier, the randomized cohort timestamp, the certificate, and the public/private key pair to software **107**. For example, trusted domain server **140** can transmit the randomized cohort, the signed certificate, and the public/private key pair to browser **107**.

(101) In some implementations, steps **4A** and **4B** occur simultaneously. In some implementations, steps **4A** and **4B** occur asynchronously.

(102) Process **350** continues with step **5**, in which software **107** provides a subsequent request to electronic doc server **104**. Because trusted domain server **140** has assigned a globally-scoped randomized cohort to software **107**, software **107** includes the randomized cohort identifier and an age bucket with its request to electronic doc server **104**. For example, electronic doc server **104** for domain ExampleTextandImageHostingPlatform.com can receive a request from browser **107** for an image as well as the assigned randomized cohort identifier and randomized cohort age bucket.

(103) Optionally, the user of software **107** can choose a verification server to independently verify the privacy and/or statistical properties of the randomized cohort identifier associated with software **107** as described with respect to steps **6**, **7**, and **8**.

(104) Process **350** continues with step **6**, in which software **107** provides its certificate to verification server **130**. For example, browser **107** can provide its signed certificate, generated by randomized cohort generator **142**, to verification server **130**.

(105) Process **350** continues with step **7**, in which verification server **130** performs the verification process based on the certificate. For example, verification server **130** can perform the verification process based on the certificate to determine statistical and/or privacy preserving properties of the randomized cohort identifier provided in the certificate.

(106) Optionally, verification server **130** can also authenticate the certificate provided prior to performing the verification process to determine that the randomized cohort identifier provided was assigned to the software **107** that is providing the certificate. For example, verification server **130** can use the public/private key pair to determine that the randomized cohort identifier provided was assigned to browser **107** that is providing the certificate. This protocol prevents the verification server **130** from considering statistics from browsers **107** with fake or stolen certificates, improving the robustness of the statistical activity data collected and reducing resources being co-opted for fraudulent use.

(107) Process **350** continues with step **8**, in which verification server **130** provides the results of the verification process to software **107**. For example, verification server **130** can output unprocessed numbers to browser **107** or a result of a statistical analysis and comparison to thresholds to determine whether statistically unusual or fraudulent activity has been detected, or whether the guaranteed level of privacy was preserved for browser **107**.

(108) FIG. **4** is a flow diagram that illustrates an example process **400** for issuing and implementing a privacy-preserving monitoring mechanism. Operations of the process **400** can be implemented, for example, by user device **106** and/or software **107**, electronic doc servers **104**, DCDS **110**, verification server **130**, trusted domain server **140**, and/or third party server **150**. Operations of the process **400** can also be implemented as instructions stored on one or more computer readable media which may be non-transitory, and execution of the instructions by one or more data processing apparatus can cause the one or more data processing apparatus to perform the operations of the process **400**.

(109) Process **400** begins with receiving, from an application on a user device of a user, a request for digital content from a domain (**402**). For example, electronic doc server **104** can receive a request for digital content from a domain associated with electronic doc server **104** from browser **107** of user device **106** as described above with respect to FIGS. **1**, **2**, and **3A-3B**.

(110) Process **400** continues with assigning, to the application and at a first time, a randomized cohort generated based on a randomly selected identifier and a timestamp indicating the first time at which the randomized cohort was assigned to the application (**404**). For example, electronic doc

server **104** can assign a locally-scoped randomized cohort associated with the domain to browser **107** as described above with respect to FIGS. **1**, **2**, and **3A**. In another example, trusted domain server **140** can assign a globally-scoped randomized cohort that can be used to browser **107** as described above with respect to FIGS. **1**, **2**, and **3B**.

(111) Process **400** continues with providing, to the application and at the first time, (i) a certificate corresponding to the randomly selected identifier and the timestamp and signed with a unique public key and (ii) a unique private key corresponding to the unique public key, where the randomly selected identifier is also assigned to at least a threshold number of other applications executing on other user devices within a predetermined period of time (**406**). For example, electronic doc server **104** or trusted domain server **140** provides the randomized cohort and/or certificate to browser **107**. If electronic doc server **104** generates a locally-scoped randomized cohort, electronic doc server **104** also provides the requested digital content to browser **107**. If trusted domain server **140** generates a globally-scoped randomized cohort, electronic doc server **104** provides the requested digital content to browser **107**. Step **406** can be performed as described above with respect to FIGS. **1**, **2**, and **3A-3B**.

(112) In some implementations, process **400** can include receiving, from the application, a second request for digital content from the domain, and providing, by the application and to the domain at a second time, a vague identifier generated based on the randomly selected identifier and a randomized cohort age bucket that indicates a range of ages of cookies that contains the age of the randomized cohort, wherein the age of the randomized cohort is calculated based on the difference between the second time and the first time. For example, the browser **107** can calculate a randomized cohort age bucket based on the randomized cohort and/or certificate provided by electronic doc server **104** or trusted domain server **140** as described above with respect to FIGS. **1**, **2**, and **3A-3B**. In some implementations, the vague identifier includes the randomly selected identifier and the randomized cohort age bucket. In some implementations, the vague identifier includes data and parameters that are derived from the randomly selected identifier and/or the randomized cohort age bucket. For example, the vague identifier can include a parameter that was derived by applying a multiplier, adding a constant, or applying some other set of operations to the randomly selected identifier and/or the randomized cohort age bucket. The vague identifier can also include data and parameters other than the randomly selected identifier and the randomized cohort age bucket.

(113) In some implementations, assigning the randomized cohort to the application includes assigning, by the domain, the randomized cohort to the application, where the randomly selected identifier is assigned to the at least a threshold number of other applications by the domain, where the randomly selected identifier is a randomly generated identifier selected from among two or more randomly generated identifiers, and where the unique public key is generated by the domain. For example, electronic doc server **104** can generate and assign the randomized cohort to the browser **107** and ensure that the randomized cohort identifier is assigned to at least $k-1$ other browsers **107** by electronic doc server **104**, where the randomized cohort identifier was selected from among two or more randomly generated identifiers, and where the unique public key is generated by electronic doc server **104** as described above with respect to FIGS. **1**, **2**, and **3A**.

(114) In some implementations, process **400** includes providing, by the application and to a verification system, the certificate, and verifying, by the verification system, that the randomized cohort is assigned to at least a threshold number of people. For example, verification server **130** can determine statistical and/or privacy properties of the randomized cohort indicated in the certificate as described above with respect to FIGS. **1**, **2**, and **3A**.

(115) In some implementations, assigning the randomized cohort to the application includes assigning, by a central server, the randomized cohort to the application, where the randomly selected identifier is assigned to the at least a threshold number of other applications, where the randomly selected identifier is a randomly generated identifier selected from among two or more

randomly generated identifiers, and where the unique public key is generated by the central server. For example, trusted domain server **140** can generate and assign the randomized cohort to the browser **107** and ensure that the randomized cohort identifier is assigned to at least $k-1$ other browsers **107**, where the randomized cohort identifier was selected from among two or more randomly generated identifiers, and where the unique public key is generated by trusted domain server **140** as described above with respect to FIGS. **1**, **2**, and **3B**.

(116) In some implementations, process **400** includes providing, by the application, a request for digital content from a second domain different from the first domain and the randomized cohort, in response to providing the request for digital content from the second domain, receiving, by the application and from the second domain, an attestation request that includes a challenge, and providing, by the application and to a verification system, the certificate, which triggers to cause the verification system to (i) create a vague certification comprising the randomly selected identifier, the randomized cohort age bucket, and the challenge, (ii) sign the vague certification, and (iii) provide the vague certification to the second domain, where the challenge is blinded from verification server using a blinding scheme. For example, verification server **130** can generate an anonymized certificate based on an attestation request as described above with respect to FIGS. **1**, **2**, and **3B**.

(117) In some implementations, process **400** includes detecting, by the domain and based on the received randomized cohort age bucket, abnormal activity associated with the randomly selected identifier and at least one of: a number of interactions associated with the randomly selected identifier, a randomized cohort age distribution, and a probability distribution associated with a particular interaction and a particular period of time. For example, analyzer **123** of electronic server **103** can detect abnormal activity associated with the randomized cohort identifier and/or a randomized cohort identifier/age bucket pair as described above with respect to FIGS. **1**, **2**, and **3A-3B**.

(118) The present technique thus allows tracking of internet activity (e.g. to help prevent abuse or fraud) whilst helping improve individual user privacy. The use of randomized cohort identifiers which are assigned to more than one user and age buckets rather than specific timestamps means the individual user associated with a given randomized cohort cannot be identified. The privacy of that user is therefore improved. At the same time, the use of randomized cohort identifier/age bucket pairs provides sufficient statistical information over time to identify abnormalities in internet activity potentially indicative of fraud. Furthermore, through the use of the verification system and challenges issued by third party domains, third party domains are able to verify a randomized cohort as genuine without the need to obtain identifiable information about an individual user. The present technique thus provides a robust way of combating abusive or fraudulent internet activity whilst helping to improve individual user privacy.

(119) FIG. **5** is a block diagram of an example computer system **500** that can be used to perform operations described above. The system **500** includes a processor **510**, a memory **520**, a storage device **530**, and an input/output device **540**. Each of the components **510**, **520**, **530**, and **540** can be interconnected, for example, using a system bus **550**. The processor **510** is capable of processing instructions for execution within the system **500**. In some implementations, the processor **510** is a single-threaded processor. In another implementation, the processor **510** is a multi-threaded processor. The processor **510** is capable of processing instructions stored in the memory **520** or on the storage device **530**.

(120) The memory **520** stores information within the system **500**. In one implementation, the memory **520** is a computer-readable medium. In some implementations, the memory **520** is a volatile memory unit. In another implementation, the memory **520** is a non-volatile memory unit.

(121) The storage device **530** is capable of providing mass storage for the system **500**. In some implementations, the storage device **530** is a computer-readable medium. In various different implementations, the storage device **530** can include, for example, a hard disk device, an optical

disk device, a storage device that is shared over a network by multiple computing devices (e.g., a cloud storage device), or some other large capacity storage device.

(122) The input/output device **540** provides input/output operations for the system **500**. In some implementations, the input/output device **540** can include one or more of a network interface devices, e.g., an Ethernet card, a serial communication device, e.g., and RS-232 port, and/or a wireless interface device, e.g., and 802.11 card. In another implementation, the input/output device can include driver devices configured to receive input data and send output data to external devices **560**, e.g., keyboard, printer and display devices. Other implementations, however, can also be used, such as mobile computing devices, mobile communication devices, set-top box television client devices, etc.

(123) Although an example processing system has been described in FIG. 4, implementations of the subject matter and the functional operations described in this specification can be implemented in other types of digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them.

(124) Embodiments of the subject matter and the operations described in this specification can be implemented in digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them. Embodiments of the subject matter described in this specification can be implemented as one or more computer programs, i.e., one or more modules of computer program instructions, encoded on computer storage media (or medium) for execution by, or to control the operation of, data processing apparatus. Alternatively, or in addition, the program instructions can be encoded on an artificially-generated propagated signal, e.g., a machine-generated electrical, optical, or electromagnetic signal, that is generated to encode information for transmission to suitable receiver apparatus for execution by a data processing apparatus. A computer storage medium can be, or be included in, a computer-readable storage device, a computer-readable storage substrate, a random or serial access memory array or device, or a combination of one or more of them. Moreover, while a computer storage medium is not a propagated signal, a computer storage medium can be a source or destination of computer program instructions encoded in an artificially-generated propagated signal. The computer storage medium can also be, or be included in, one or more separate physical components or media (e.g., multiple CDs, disks, or other storage devices).

(125) The operations described in this specification can be implemented as operations performed by a data processing apparatus on data stored on one or more computer-readable storage devices or received from other sources.

(126) The term “data processing apparatus” encompasses all kinds of apparatus, devices, and machines for processing data, including by way of example a programmable processor, a computer, a system on a chip, or multiple ones, or combinations, of the foregoing. The apparatus can include special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application-specific integrated circuit). The apparatus can also include, in addition to hardware, code that creates an execution environment for the computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, a cross-platform runtime environment, a virtual machine, or a combination of one or more of them. The apparatus and execution environment can realize various different computing model infrastructures, such as web services, distributed computing and grid computing infrastructures.

(127) A computer program (also known as a program, software, software application, script, or code) can be written in any form of programming language, including compiled or interpreted languages, declarative or procedural languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, object, or other unit suitable for use in a computing environment. A computer program may, but need not, correspond to a file in a file

system. A program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub-programs, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

(128) The processes and logic flows described in this specification can be performed by one or more programmable processors executing one or more computer programs to perform actions by operating on input data and generating output. The processes and logic flows can also be performed by, and apparatus can also be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application-specific integrated circuit).

(129) Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors. Generally, a processor will receive instructions and data from a read-only memory or a random access memory or both. The essential elements of a computer are a processor for performing actions in accordance with instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto-optical disks, or optical disks. However, a computer need not have such devices. Moreover, a computer can be embedded in another device, e.g., a mobile telephone, a personal digital assistant (PDA), a mobile audio or video player, a game console, a Global Positioning System (GPS) receiver, or a portable storage device (e.g., a universal serial bus (USB) flash drive), to name just a few. Devices suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

(130) To provide for interaction with a user, embodiments of the subject matter described in this specification can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube) or LCD (liquid crystal display) monitor, for displaying information to the user and a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input. In addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; for example, by sending web pages to a web browser on a user's client device in response to requests received from the web browser.

(131) Embodiments of the subject matter described in this specification can be implemented in a computing system that includes a back-end component, e.g., as a data server, or that includes a middleware component, e.g., an application server, or that includes a front-end component, e.g., a client computer having a graphical user interface or a Web browser through which a user can interact with an implementation of the subject matter described in this specification, or any combination of one or more such back-end, middleware, or front-end components. The components of the system can be interconnected by any form or medium of digital data communication, e.g., a communication network. Examples of communication networks include a local area network ("LAN") and a wide area network ("WAN"), an inter-network (e.g., the Internet), and peer-to-peer networks (e.g., ad hoc peer-to-peer networks).

(132) The computing system can include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship

of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other. In some embodiments, a server transmits data (e.g., an HTML page) to a client device (e.g., for purposes of displaying data to and receiving user input from a user interacting with the client device). Data generated at the client device (e.g., a result of the user interaction) can be received from the client device at the server.

(133) While this specification contains many specific implementation details, these should not be construed as limitations on the scope of any inventions or of what may be claimed, but rather as descriptions of features specific to particular embodiments of particular inventions. Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

(134) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

(135) Thus, particular embodiments of the subject matter have been described. Other embodiments are within the scope of the following claims. In some cases, the actions recited in the claims can be performed in a different order and still achieve desirable results. In addition, the processes depicted in the accompanying figures do not necessarily require the particular order shown, or sequential order, to achieve desirable results. In certain implementations, multitasking and parallel processing may be advantageous.

Claims

1. A method for privacy-preserving web activity monitoring, comprising: receiving, from an application on a user device of a user, a request for digital content from a domain; assigning, to the application and at a first time, a randomized cohort constructed based on (i) a randomly selected identifier and (ii) a timestamp indicating the first time at which the randomized cohort was assigned to the application, wherein the randomly selected identifier is also assigned to at least a threshold number of other applications executing on other user devices within a predetermined period of time of the assignment of the randomly selected identifier to the application; providing, to the application and at the first time, (i) a digitally signed certificate corresponding to the randomly selected identifier and the timestamp and (ii) a unique public key and corresponding unique private key associated with the certificate; receiving, from the application and at a second time, (i) a second request for digital content from the domain and (ii) a vague identifier corresponding to the randomly selected identifier and a randomized cohort age bucket that indicates a range of ages of cookies that contains the age of the randomized cohort, wherein the age of the randomized cohort is calculated based on a difference between the second time and the first time, and wherein the randomized cohort age bucket obfuscates the age of the randomized cohort; receiving, from the application and after the second time, subsequent requests for digital content from the domain, wherein each subsequent request includes the vague identifier to anonymize the user; and providing, to the application, the requested digital content in response to each subsequent request.

2. The method of claim 1, wherein assigning, to the application, a randomized cohort comprises: assigning, by the domain, the randomized cohort to the application, and wherein the randomly selected identifier is assigned to the at least a threshold number of other applications, wherein the randomly selected identifier is a randomly generated identifier selected from among two or more randomly generated identifiers, and wherein the unique public key is generated by the domain.
3. The method of claim 1, wherein assigning, to the application, a randomized cohort comprises: assigning, by a central server, the randomized cohort to the application, wherein the randomly selected identifier is assigned to the at least a threshold number of other applications, wherein the randomly selected identifier is a randomly generated identifier selected from among two or more randomly generated identifiers, and wherein the unique public key is generated by the central server.
4. The method of claim 1, further comprising: providing, by the application, a request for digital content from a second domain different from the domain and the randomized cohort; in response to providing the request for digital content from the second domain, receiving, by the application and from the second domain, an attestation request that includes a challenge; and providing, by the application and to a verification system, the digitally signed certificate, which triggers the verification system to (i) create a vague certification comprising the randomly selected identifier, the randomized cohort age bucket, and the challenge, (ii) sign the vague certification, and (iii) provide the vague certification to the second domain, wherein the challenge is blinded from the verification system using a blinding scheme.
5. The method of claim 1, further comprising: providing, by the application and to a verification system, the digitally signed certificate; and verifying, by the verification system, that the randomly selected identifier is assigned to at least the threshold number of applications.
6. The method of claim 1, further comprising: detecting, by the domain and based on the randomized cohort age bucket, abnormal activity associated with the randomly selected identifier and at least one of: a number of interactions associated with the randomly selected identifier, a randomized cohort age distribution, and a probability distribution associated with a particular interaction and a particular period of time.
7. A system comprising: one or more processors; and one or more memory elements including instructions that, when executed, cause the one or more processors to perform operations including: receiving, from an application on a user device of a user, a request for digital content from a domain; assigning, to the application and at a first time, a randomized cohort constructed based on (i) a randomly selected identifier and (ii) a timestamp indicating the first time at which the randomized cohort was assigned to the application, wherein the randomly selected identifier is also assigned to at least a threshold number of other applications executing on other user devices within a predetermined period of time of the assignment of the randomly selected identifier to the application; providing, to the application and at the first time, (i) a digitally signed certificate corresponding to the randomly selected identifier and the timestamp and (ii) a unique public key and corresponding unique private key associated with the certificate; receiving, from the application and at a second time, (i) a second request for digital content from the domain and (ii) a vague identifier corresponding to the randomly selected identifier and a randomized cohort age bucket that indicates a range of ages of cookies that contains the age of the randomized cohort, wherein the age of the randomized cohort is calculated based on a difference between the second time and the first time, and wherein the randomized cohort age bucket obfuscates the age of the randomized cohort; receiving, from the application and after the second time, subsequent requests for digital content from the domain, wherein each subsequent request includes the vague identifier to anonymize the user; and providing, to the application, the requested digital content in response to each subsequent request.
8. The system of claim 7, wherein assigning, to the application, a randomized cohort comprises: assigning, by the domain, the randomized cohort to the application, and wherein the randomly

selected identifier is assigned to the at least a threshold number of other applications, wherein the randomly selected identifier is a randomly generated identifier selected from among two or more randomly generated identifiers, and wherein the unique public key is generated by the domain.

9. The system of claim 7, wherein assigning, to the application, a randomized cohort comprises: assigning, by a central server, the randomized cohort to the application, wherein the randomly selected identifier is assigned to the at least a threshold number of other applications, wherein the randomly selected identifier is a randomly generated identifier selected from among two or more randomly generated identifiers, and wherein the unique public key is generated by the central server.

10. The system of claim 7, the operations further comprising: providing, by the application, a request for digital content from a second domain different from the domain and the randomized cohort; in response to providing the request for digital content from the second domain, receiving, by the application and from the second domain, an attestation request that includes a challenge; and providing, by the application and to a verification system, the digitally signed certificate, which triggers the verification system to (i) create a vague certification comprising the randomly selected identifier, the randomized cohort age bucket, and the challenge, (ii) sign the vague certification, and (iii) provide the vague certification to the second domain, wherein the challenge is blinded from the verification system using a blinding scheme.

11. The system of claim 7, the operations further comprising: providing, by the application and to a verification system, the digitally signed certificate; and verifying, by the verification system, that the randomly selected identifier is assigned to at least the threshold number of applications.

12. The system of claim 7, the operations further comprising: detecting, by the domain and based on the randomized cohort age bucket, abnormal activity associated with the randomly selected identifier and at least one of: a number of interactions associated with the randomly selected identifier, a randomized cohort age distribution, and a probability distribution associated with a particular interaction and a particular period of time.

13. A non-transitory computer storage medium encoded with instructions that when executed by a distributed computing system cause the distributed computing system to perform operations comprising: receiving, from an application on a user device of a user, a request for digital content from a domain; assigning, to the application and at a first time, a randomized cohort constructed based on (i) a randomly selected identifier and (ii) a timestamp indicating the first time at which the randomized cohort was assigned to the application, wherein the randomly selected identifier is also assigned to at least a threshold number of other applications executing on other user devices within a predetermined period of time of the assignment of the randomly selected identifier to the application; providing, to the application and at the first time, (i) a digitally signed certificate corresponding to the randomly selected identifier and the timestamp and (ii) a unique public key and corresponding unique private key associated with the certificate; receiving, from the application and at a second time, (i) a second request for digital content from the domain and (ii) a vague identifier corresponding to the randomly selected identifier and a randomized cohort age bucket that indicates a range of ages of cookies that contains the age of the randomized cohort, wherein the age of the randomized cohort is calculated based on a difference between the second time and the first time, and wherein the randomized cohort age bucket obfuscates the age of the randomized cohort; receiving, from the application and after the second time, subsequent requests for digital content from the domain, wherein each subsequent request includes the vague identifier to anonymize the user; and providing, to the application, the requested digital content in response to each subsequent request.

14. The non-transitory computer storage medium of claim 13, wherein assigning, to the application, a randomized cohort comprises: assigning, by the domain, the randomized cohort to the application, and wherein the randomly selected identifier is assigned to the at least a threshold number of other applications, wherein the randomly selected identifier is a randomly generated

identifier selected from among two or more randomly generated identifiers, and wherein the unique public key is generated by the domain.

15. The non-transitory computer storage medium of claim 13, wherein assigning, to the application, a randomized cohort comprises: assigning, by a central server, the randomized cohort to the application, wherein the randomly selected identifier is assigned to the at least a threshold number of other applications, wherein the randomly selected identifier is a randomly generated identifier selected from among two or more randomly generated identifiers, and wherein the unique public key is generated by the central server.

16. The non-transitory computer storage medium of claim 13, the operations further comprising: providing, by the application, a request for digital content from a second domain different from the domain and the randomized cohort; in response to providing the request for digital content from the second domain, receiving, by the application and from the second domain, an attestation request that includes a challenge; and providing, by the application and to a verification system, the digitally signed certificate, which triggers the verification system to (i) create a vague certification comprising the randomly selected identifier, the randomized cohort age bucket, and the challenge, (ii) sign the vague certification, and (iii) provide the vague certification to the second domain, wherein the challenge is blinded from the verification system using a blinding scheme.

17. The non-transitory computer storage medium of claim 13, the operations further comprising: providing, by the application and to a verification system, the digitally signed certificate; and verifying, by the verification system, that the randomly selected identifier is assigned to at least the threshold number of applications.

18. The non-transitory computer storage medium of claim 13, wherein the operations comprise: detecting, by the domain and based on the randomized cohort age bucket, abnormal activity associated with the randomly selected identifier and at least one of: a number of interactions associated with the randomly selected identifier, a randomized cohort age distribution, and a probability distribution associated with a particular interaction and a particular period of time.
